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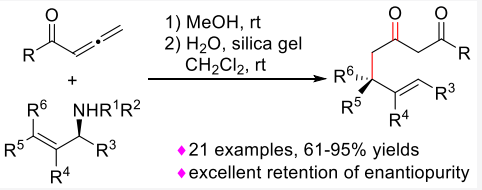
Asymmetric Aza-Claisen Rearrangement between Enantioenriched α-Chiral Allylamines and Allenones

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ABSTRACT: An unprecedented asymmetric aza-Claisen rearrangement between enantioenriched α-chiral allylamines and allenones was found to proceed in the absence of catalysts and additives at room temperature. The rearrangement, followed by hydrolysis, provides convenient access to structurally diverse δ-chiral β-diketones in good to excellent yields with excellent retention of enantiopurity. This protocol proved powerful for the construction of an all-carbon quaternary stereocenter with high enantiopurity.

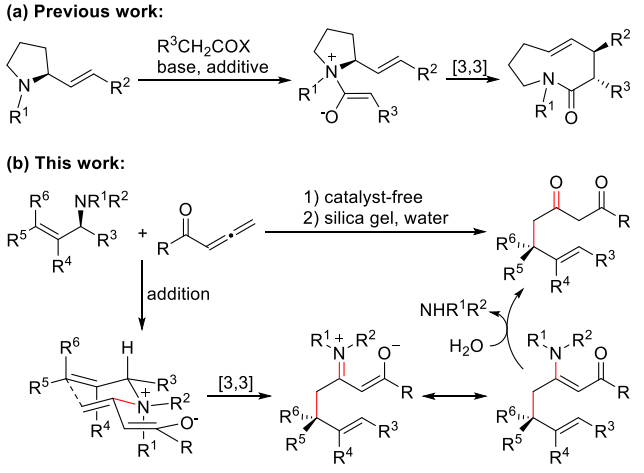


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The aza-Claisen rearrangement of allylamines is a versatile C−N bond cleavage reaction that serves as a key step in the construction of complex nitrogen-containing molecules.[1] Charge acceleration has been frequently employed as a powerful strategy to decrease the reaction temperature via converting allylamines into the corresponding ammonium salts through protonation, Lewis acid coordination, or quaternization prior to the aza-Claisen rearrangement. Particularly, zwitterionic aza-Claisen rearrangements can proceed under relatively mild conditions after nucleophilic addition of tertiary allylamines to π-electrophiles such as 2-alkynoates,[2] acetylenic sulfones,[3] allenoates,[4] ketenes,[5,6] ketenimines,[7] carbodiimides, isocyanates, and isothiocyanates.[8] While stereochemical advantages can be expected from the aza-Claisen rearrangements because of their chair-like six-membered transition states with minimized repulsive interactions, only L-(−)-proline-derived α-chiral allylamines have been reported previously to exhibit effective 1,3-chirality transfer in the ketene-mediated rearrangement/ring-expansion reaction (Scheme 1a).[5d,e,g] To our knowledge, 1,3-chirality transfer has not yet been disclosed for the aza-Claisen rearrangement of other types of enantioenriched α-chiral allylamines, which, though, can be readily accessible by resolution of the racemic ones with inexpensive chiral acids.[9] In connection with our exploration of new reactions of electrophile-activated tertiary amines,[10] we have developed an unprecedented asymmetric zwitterionic aza-Claisen rearrangement between enantioenriched α-chiral allylamines and allenones in the absence of catalysts and additives at room temperature (Scheme 1b). Importantly, the rearrangement, followed by hydrolysis, provides convenient access to structurally diverse δ-chiral β-diketones with excellent retention of enantiopurity.

Inspired by previous reports on the zwitterionic aza-Claisen rearrangements of π-electrophile-activated tertiary allylamines,[2−8] we initiated our investigation on a possible asymmetric aza-Claisen rearrangement between enantioenriched α-chiral allylamines and allenones,[11] wherein retention of enantiopurity was expected via a chair-like six-membered transition state with minimized repulsive interactions (Scheme 1b). To our delight, simply treating allylamine ***rac*-1a** with allenone **2a** in dichloromethane at room temperature afforded β-amino enone ***rac*-3a** in 85% NMR yield (Table 1, entry 1). It is noteworthy that the disubstituted alkene moiety was formed with exclusive *E* selectivity, which is in line with the mechanism involving a privileged chair-like six-membered transition state as shown in Scheme 1b. We next screened a

Scheme 1. 1,3-Chirality Transfer in the Zwitterionic Aza-Claisen Rearrangements of Enantioenriched α-Chiral Allylamines



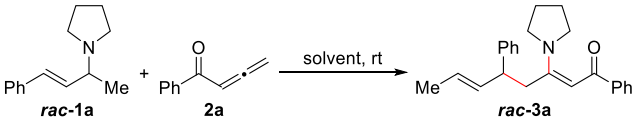
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Table 1. Screening of Solvents[a]



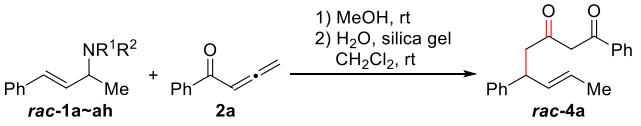
entry	solvent	yield (%)[b]
1	CH_2Cl_2	85
2	$CHCl_3$	82
3	DCE	50
4	PhMe	68
5	hexane	45
6	EtOAc	61
7	THF	70
8	dioxane	60
9	acetone	94
10	MeCN	65
11	DMF	92
12	DMSO	89
13	MeOH	98
14	EtOH	83

[a]Reaction conditions: ***rac*-1a** (0.20 mmol), **2a** (0.24 mmol), solvent (1.0 mL), rt, 12 h. [b]Determined by ^{1}H NMR spectroscopic analysis of the crude product.

number of common solvents and found that the use of methanol improved the NMR yield to 98% (Table 1, entry 13).

Since β-amino enone ***rac*-3a** was hydrolyzed partially when being purified by silica gel chromatography, we removed methanol from the reaction mixture of allylamine ***rac*-1a** and allenone **2a** under reduced pressure at room temperature and then treated the residue with silica gel in dichloromethane, which contained 1.0% (v/v) water. This protocol afforded δ-branched β-diketone ***rac*-4a** in 95% yield (Table 2, entry 1).

Table 2. Survey of the *N*-Substituents[a]



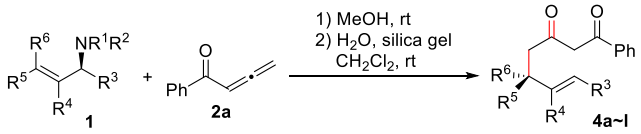
entry	amine	R^1	R^2	yield (%)[b]
1	***rac*-1a**	$(CH_2)_4$		95
2	***rac*-1ab**	$(CH_2)_5$		85
3	***rac*-1ac**	$(CH_2)_2O(CH_2)_2$		82
4	***rac*-1ad**	Me	Me	90
5	***rac*-1ae**	Me	CH_2Ph	93
6	***rac*-1af**	Me	Ph	34
7	***rac*-1ag**	Me	H	0
8	***rac*-1ah**	H	H	0

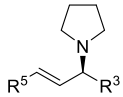
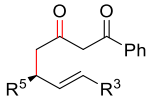
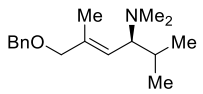
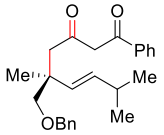
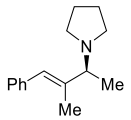
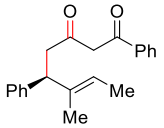
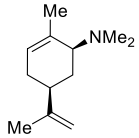
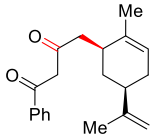
[a]Reaction conditions: (1) ***rac*-1a–ah** (0.20 mmol), **2a** (0.24 mmol), methanol (1.0 mL), rt, 12 h; (2) silica gel (500 mg), dichloromethane (5.0 mL, containing 1.0% (v/v) water), rt, 12 h. [b]Isolated yield.

Next, we surveyed the effects of the *N*-substituent in the allylamine on the aza-Claisen rearrangement. In general, the reaction with an *N*,*N*-dialkyl allylamine gave a much better yield than that with a less nucleophilic allylamine bearing an *N*-aryl group (Table 2, entries 1–6). Not surprisingly, the aza-Claisen rearrangement was not observed in the corresponding reaction of either secondary allylamine ***rac*-1ag** or primary allylamine ***rac*-1ah** under the standard conditions (Table 2, entries 7 and 8).

Encouraged by the above results, we examined an asymmetric version of the aza-Claisen rearrangement. Treatment of enantioenriched α-chiral allylamine **1a** (97% ee) with allenone **2a**, followed by hydrolysis at room temperature, afforded δ-chiral β-diketone **4a** with complete retention of enantiopurity (Table 3, entry 1). The protocol for sequential

Table 3. Scope of Enantioenriched α-Chiral Allylamines[a]



entry	amine **1**	product **4**
		
1	**1a**, R^3 = Me, R^5 = Ph (97% ee)	**4a**, 95%, 97% ee
2	**1b**, R^3 = Et, R^5 = Ph (97% ee)	**4b**, 83%, 97% ee
3	**1c**, R^3 = $(CH_2)_2Ph$, R^5 = Ph (96% ee)	**4c**, 90%, 96% ee
4	**1d**, R^3 = ^{i}Pr, R^5 = Ph (98% ee)	**4d**, 90%, 98% ee
5	**1e**, R^3 = R^5 = Ph (96% ee)	**4e**, 78%, 96% ee
6	**1f**, R^3 = Me, R^5 = 2-ClC_6H_4 (99% ee)	**4f**, 90%, 99% ee
7	**1g**, R^3 = Et, R^5 = 4-$MeOC_6H_4$ (80% ee)	**4g**, 90%, 80% ee
8	**1h**, R^3 = Me, R^5 = (*E*)-styryl (99% ee)	**4h**, 89%, 99% ee
9	**1i**, R^3 = Me, R^5 = Cy (97% ee)	**4i**, 88%, 97% ee
		
10	**1j** (>99% ee)	**4j**, 87%, 95% ee
		
11	**1k** (99% ee)	**4k**, 85%, 99% ee
		
12	**1l** (95:5 dr)	**4l**, 91%, 93:7 dr

[a]Reaction conditions: (1) **1** (0.20 mmol), **2a** (0.24 mmol), methanol (1.0 mL), rt, 12 h; (2) silica gel (500 mg), dichloromethane (5.0 mL, containing 1.0% (v/v) water), rt, 12 h. Isolated yields are given. The ee values were determined by HPLC analysis on a chiral stationary phase. The dr values were determined by ^{1}H NMR spectroscopic analysis of the crude products.

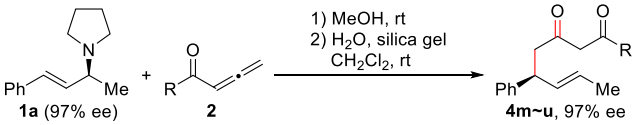
aza-Claisen rearrangement and hydrolysis was successfully extended to a variety of enantioenriched polysubstituted α-chiral allylamines, and structurally diverse δ-chiral β-diketones were obtained in good to excellent yields with excellent retention of enantiopurity (Table 3, entries 2–11). Notably, in these cases, the di- and trisubstituted alkene moieties were

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formed with exclusive *E* selectivity. Moreover, the reaction proved powerful for the construction of an all-carbon quaternary stereocenter with high enantiopurity (Table 3, entry 10). We also examined (*S*)-(+)-carvone-derived α-chiral allylamine **1l** and obtained δ-chiral β-diketone **4l** with excellent diastereoselectivity (Table 3, entry 12).

On the other hand, the aza-Claisen rearrangement worked well with a range of allenones bearing a monosubstituted allenyl group, including 1-aryl-, 1-heteroaryl-, 1-alkenyl-, and 1-alkylbuta-2,3-dien-1-ones (Table 4). In these cases, the

Table 4. Scope of Allenones[a]



entry	allenone 2	R	product 4	yield (%)[b]
1	**2b**	4-MeOC$_6$H$_4$	**4m**	88
2	**2c**	4-FC$_6$H$_4$	**4n**	79
3	**2d**	4-BrC$_6$H$_4$	**4o**	89
4	**2e**	2-ClC$_6$H$_4$	**4p**	80
5	**2f**	2,4,6-Me$_3$C$_6$H$_2$	**4q**	61
6	**2g**	2-furyl	**4r**	83
7	**2h**	2-thienyl	**4s**	83
8	**2i**	(*E*)-styryl	**4t**	75
9	**2j**	CH$_2$Ph	**4u**	82

[a]Reaction conditions: (1) **1a** (0.20 mmol), **2** (0.24 mmol), methanol (1.0 mL), rt, 12 h; (2) silica gel (500 mg), dichloromethane (5.0 mL, containing 1.0% (v/v) water), rt, 12 h. The ee values were determined by HPLC analysis on a chiral stationary phase. The absolute configuration of product **4o** was assigned to be *S* by X-ray crystallography (CCDC 1489729), and that of the rest of the products (shown in Tables 3 and 4) was assigned by analogy. [b]Isolated yield.

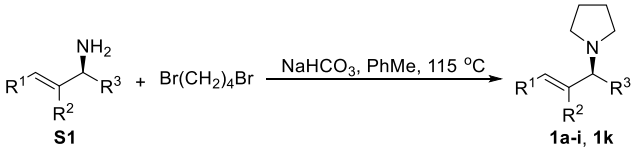
reaction exhibited complete retention of enantiopurity and the alkene moieties were formed with exclusive *E* selectivity. Nevertheless, the reaction did not occur with bulkier allenones such as 2-methyl-1-phenylbuta-2,3-dien-1-one (**2k**) and 4-methyl-1-phenylpenta-2,3-dien-1-one (**2l**). Moreover, the absolute configuration of δ-chiral β-diketone **4o** was assigned to be *S* by X-ray crystallography and the stereochemistry for its formation is in line with the mechanism of 1,3-chirality transfer involving a chair-like six-membered transition state as depicted in Scheme 1b.

In summary, we have developed an efficient asymmetric aza-Claisen rearrangement between enantioenriched α-chiral allylamines and allenones in the absence of catalysts and additives at room temperature. The rearrangement, followed by hydrolysis, provides convenient access to structurally diverse δ-chiral β-diketones in good to excellent yields with excellent retention of enantiopurity. In these cases, the di- and trisubstituted alkene moieties were formed with exclusive *E* selectivity. Moreover, this protocol proved powerful for the construction of an all-carbon quaternary stereocenter with high enantiopurity.

EXPERIMENTAL SECTION

General Information. ^{1}H and ^{13}C{^{1}H} NMR spectra were recorded on 400 and 100 MHz spectrometers, or on 500 and 125 MHz spectrometers, respectively, and were referenced internally with tetramethylsilane (δ H 0.00), CDCl$_3$ (δ C 77.2). NMR multiplicities were abbreviated as follows: s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, br = broad. Chemical shifts (δ) and coupling constants (J) were expressed in ppm and Hz, respectively. High resolution mass spectra (HRMS) were recorded on an LC-TOF spectrometer using electron spray ionization (ESI) techniques. High pressure liquid chromatography (HPLC) analyses were performed on an instrument equipped with an isostatic pump using a chiral column (AD-H, OJ, OJ-H, OD-H, IC, 250 × 4.6 mm) with isopropanol/hexane as mobile phase, and the UV detection was monitored at 254 or 220 nm. The chiral HPLC methods were calibrated with the corresponding racemic mixtures. Optical rotations were measured on a polarimeter with a sodium lamp at λ = 589 nm and reported as $[\alpha]_D^{T°C}$ (c = g/100 mL, solvent). Melting points are uncorrected. Racemic amines **1a**, **1b**, **1e**, and **1ab**–**ah**[9d,12] and allenones **2a**–**l**[13] were prepared according to literature procedures. Abbreviations: Ac = acetyl, Bn = benzyl, Boc = *tert*-butoxycarbonyl, Cy = cyclohexyl, DCE = 1,2-dichloroethane, DMF = *N*,*N*-dimethylformamide, DMSO = dimethyl sulfoxide, rt = room temperature, THF = tetrahydrofuran, TLC = thin layer chromatography.

Preparation of Allylamines 1a–i and 1k.[9g]



Chiral primary allylamines **S1** were prepared according to literature procedures.[9a−d,14]

A mixture of chiral primary allylamine **S1** (1.0 mmol), 1,4-dibromobutane (238 mg, 1.1 mmol), and sodium bicarbonate (185 mg, 2.2 mmol) in toluene (1.0 mL) was heated under nitrogen at 115 °C for 23 h. The mixture was cooled to room temperature and purified by silica gel chromatography, eluting with dichloromethane/methanol (15:1), to give the corresponding tertiary amine.

*(S,E)-1-(4-Phenylbut-3-en-2-yl)pyrrolidine (**1a**).* Yellow oil (154 mg, 76% yield); 97% ee; $[\alpha]_D^{20}$ = −14.6 (c = 0.31, CHCl$_3$); Lit.: $[\alpha]_D^{26}$ = −76.0 (c = 1.0, CHCl$_3$, 91:9 e.r.);[15] ^{1}H NMR (400 MHz, CDCl$_3$) δ 7.37 (d, J = 7.2 Hz, 2H), 7.30−7.28 (m, 2H), 7.23−7.19 (m, 1H), 6.47 (d, J = 16.0 Hz, 1H), 6.24 (dd, J = 16.0, 8.4 Hz, 1H), 2.93−2.87 (m, 1H), 2.62−2.53 (m, 4H), 1.84−1.74 (m, 4H), 1.30 (d, J = 6.4 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, CDCl$_3$) δ 137.3, 134.1, 129.8, 128.6, 127.4, 126.4, 63.2, 52.3, 23.5, 21.1; HRMS (ESI) m/z: (M + H)$^+$ Calcd for C$_{14}$H$_{20}$N 202.1590; Found 202.1588.

*(S,E)-1-(1-Phenylpent-1-en-3-yl)pyrrolidine (**1b**).* Yellow oil (148 mg, 69% yield); 97% ee; $[\alpha]_D^{20}$ = −25.9 (c = 0.54, CHCl$_3$); ^{1}H NMR (400 MHz, CDCl$_3$) δ 7.38 (d, J = 7.2 Hz, 2H), 7.32−7.29 (m, 2H), 7.24−7.20 (m, 1H), 6.46 (d, J = 16.0 Hz, 1H), 6.13 (dd, J = 16.0, 9.2 Hz, 1H), 2.68−2.54 (m, 5H), 1.89−1.73 (m, 5H),1.58−1.47 (m, 1H), 0.89 (t, J = 7.6 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, CDCl$_3$) δ 137.3, 132.1, 131.6, 128.7, 127.4, 126.4, 70.1, 52.2, 27.1, 23.3, 10.8; HRMS (ESI) m/z: (M + H)$^+$ Calcd for C$_{15}$H$_{22}$N 216.1747; Found 216.1757.

*(S,E)-1-(1,5-Diphenylpent-1-en-3-yl)pyrrolidine (**1c**).* Yellow oil (120 mg, 41% yield); 96% ee; $[\alpha]_D^{20}$ = −18.2 (c = 0.33, CHCl$_3$); ^{1}H NMR (400 MHz, CDCl$_3$) δ 7.42−7.40 (m, 2H), 7.35−7.29 (m, 2H), 7.30−7.22 (m, 3H), 7.22−7.15 (m, 3H), 6.49 (d, J = 16.0 Hz, 1H), 6.22 (dd, J = 16.0, 9.2 Hz, 1H), 2.89−2.83 (m, 1H), 2.77−2.69 (m, 1H), 2.62−2.52 (m, 5H), 2.18−2.10 (m, 1H), 1.93−1.83 (m, 1H), 1.79−1.73 (m, 4H); ^{13}C{^{1}H} NMR (100 MHz, CDCl$_3$) δ 142.4, 137.1, 132.4, 131.3, 128.7, 128.5, 128.5, 127.6, 126.5, 125.9, 67.6, 51.9, 35.9, 32.5, 23.4; HRMS (ESI) m/z: (M + H)$^+$ Calcd for C$_{21}$H$_{26}$N 292.2060; Found 292.2067.

*(S,E)-1-(4-Methyl-1-phenylpent-1-en-3-yl)pyrrolidine (**1d**).* Yellow oil (118 mg, 51% yield); 98% ee; $[\alpha]_D^{20}$ = −23.8 (c = 0.41, CHCl$_3$); ^{1}H NMR (400 MHz, CDCl$_3$) δ 7.40−7.37 (m, 2H), 7.33−7.30 (m, 2H), 7.24−7.20 (m, 1H), 6.43 (d, J = 16.0 Hz, 1H), 6.19 (dd, J = 16.0, 9.2 Hz, 1H), 2.55 (s, 5H), 2.08−2.00 (m, 1H), 1.78−1.75 (m, 4H), 0.96 (d, J = 6.8 Hz, 3H), 0.89 (d, J = 6.8 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, CDCl$_3$) δ 137.4, 132.4, 128.7, 127.4, 126.4, 74.1,

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52.2, 30.6, 23.3, 20.7, 16.8; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{16}H_{24}N$ 230.1903; Found 230.1911.

(R,E)-1-(1,3-Diphenylallyl)pyrrolidine (**1e**). Yellow solid (105 mg, 40% yield); m.p. 78−80 °C; 96% ee; $[\alpha]_D^{20}$ = +12.4 (c = 0.19, $CHCl_3$); Lit.: $[\alpha]_D^{25}$ = +13.75 (c = 0.28, $CHCl_3$, *R*-enantiomer, 92% ee);[16] ^1H NMR (400 MHz, $CDCl_3$) δ 7.42 (d, J = 7.6 Hz, 2H), 7.35−7.16 (m, 8H), 6.56 (d, J = 15.6 Hz, 1H), 6.42 (dd, J = 15.6, 8.4 Hz, 1H), 3.76 (d, J = 8.4 Hz, 1H), 2.59−2.53 (m, 2H), 2.46−2.41 (m, 2H),1.81−1.73 (m, 4H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 143.1, 137.1, 133.1, 130.0, 128.7, 128.6, 127.8, 127.5, 127.3, 126.5, 74.5, 53.3, 23.5; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{19}H_{22}N$ 264.1747; Found 264.1746.

(S,E)-1-(4-(2-Chlorophenyl)but-3-en-2-yl)pyrrolidine (**1f**). Yellow oil (102 mg, 43% yield); 99% ee; $[\alpha]_D^{20}$ = −13.8 (c = 0.19, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.54−7.52 (m, 1H), 7.35−7.32 (m, 1H), 7.23−7.13 (m, 2H), 6.86 (d, J = 15.6 Hz, 1H), 6.22 (dd, J = 15.6, 8.8 Hz, 1H), 3.00−2.93 (m, 1H), 2.65−2.53 (m, 4H), 1.85−1.73 (m, 4H), 1.31 (d, J = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 136.9, 135.4, 133.0, 129.7, 128.4, 126.9, 126.9, 126.0, 63.2, 52.4, 23.5, 21.1; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{14}H_{19}ClN$ 236.1201; Found 236.1208.

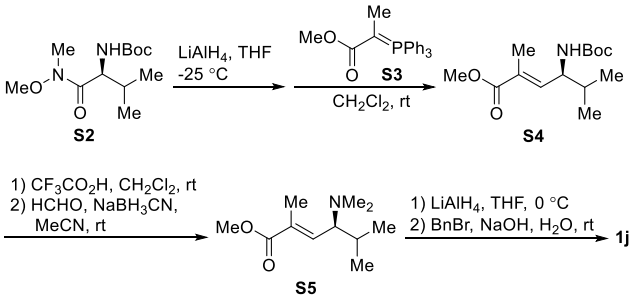
(S,E)-1-(1-(4-Methoxyphenyl)pent-1-en-3-yl)pyrrolidine (**1g**). Yellow oil (99.0 mg, 40% yield); 80% ee; $[\alpha]_D^{20}$ = −18.2 (c = 0.31, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.34−7.30 (m, 2H), 6.87−6.84 (m, 2H), 6.40 (d, J = 16.0 Hz, 1H), 5.98 (dd, J = 16.0, 8.8 Hz, 1H), 3.81 (s, 3H), 2.67−2.54 (m, 5H), 1.88−1.73 (m, 5H), 1.57−1.45 (m, 1H), 0.88 (t, J = 7.6 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 159.1, 131.1, 130.1, 129.7, 127.5, 114.1, 70.2, 55.5, 52.2, 27.2, 23.3, 10.8; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{16}H_{24}NO$ 246.1852; Found 246.1852.

1-((S,3E,5E)-6-Phenylhexa-3,5-dien-2-yl)pyrrolidine (**1h**). Yellow oil (120 mg, 53% yield); 99% ee; $[\alpha]_D^{20}$ = −16.9 (c = 0.29, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.38−7.36 (m, 2H), 7.31−7.26 (m, 2H), 7.21−7.19 (m, 1H), 6.50 (d, J = 15.6 Hz, 1H), 6.28 (dd, J = 16.0, 8.4 Hz, 1H), 5.68−5.55 (m, 2H), 3.32−3.26 (m, 1H), 2.57−2.54 (m, 4H), 1.79−1.77 (m, 4H), 1.71 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 137.3, 132.2, 131.6, 130.4, 128.6, 127.5, 127.1, 126.4, 71.2, 52.4, 23.4, 18.0; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{16}H_{22}N$ 228.1747; Found 228.1747.

(S,E)-1-(4-Cyclohexylbut-3-en-2-yl)pyrrolidine (**1i**). Light yellow oil (60.0 mg, 29% yield); 97% ee; $[\alpha]_D^{20}$ = −23.5 (c = 0.19, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 5.47 (dd, J = 15.5, 6.5 Hz, 1H), 5.37 (dd, J = 15.5, 7.5 Hz, 1H), 2.70−2.65 (m, 1H), 2.54−2.49 (m, 4H), 1.95−1.90 (m, 1H), 1.78−1.75 (m, 4H), 1.72−1.65 (m, 4H), 1.65−1.62 (m, 1H), 1.32−1.01 (m, 8H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 137.1, 131.0, 63.1, 52.2, 40.4, 33.1, 33.1, 26.3, 26.2, 23.5, 21.2; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{14}H_{26}N$ 208.2060; Found 208.2063.

(S,E)-1-(3-Methyl-4-phenylbut-3-en-2-yl)pyrrolidine (**1k**). Light yellow oil (92.0 mg, 43% yield); 99% ee; $[\alpha]_D^{20}$ = −21.2 (c = 0.36, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.34−7.30 (m, 2H), 7.28−7.25 (m, 2H), 7.22−7.17 (m, 1H), 6.44 (s, 1H), 2.77 (q, J = 6.4 Hz, 1H), 2.65−2.58 (m, 2H), 2.53−2.46 (m, 2H), 1.91 (d, J = 1.2 Hz, 3H), 1.85−1.78 (m, 4H), 1.30 (d, J = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 142.3, 138.2, 129.1, 128.2, 126.2, 125.6, 70.4, 53.0, 23.6, 20.0, 13.5; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{15}H_{22}N$ 216.1747; Found 216.1746.

Preparation of Allylamine 1j.[17,18]



To a solution of amide **S2** (1.30 g, 5.0 mmol) in dry tetrahydrofuran (25 mL) under a nitrogen atmosphere at 0 °C was added dropwise a solution of $LiAlH_4$ in tetrahydrofuran (4.0 M, 0.63 mL, 2.5 mmol). The mixture was stirred at 0 °C for 30 min and then cooled to −15°C, and a saturated aqueous $KHSO_4$ solution (15 mL) was added carefully. The solution was diluted with ether (40 mL) and stirred vigorously for 20 min. The organic layer was washed with brine, dried over anhydrous sodium sulfate, filtered, and concentrated to give the crude aldehyde as a colorless oil, which was used directly in the next step without further purification.

The crude aldehyde was dissolved in dry dichloromethane (20 mL) at room temperature, and ylide **S3** (1.92 g, 5.5 mmol) was added in one portion under a nitrogen atmosphere. After stirring for 12 h, dichloromethane (20 mL) was added and the mixture was successively washed with a 10% aqueous $KHSO_4$ solution (20 mL), saturated aqueous $NaHCO_3$ solution (20 mL), and brine. The organic layer was dried over anhydrous sodium sulfate, filtered, and concentrated. The residue was purified by column chromatography (petroleum ether:ethyl acetate = 5:1) to give compound *methyl (S,E)-4-((tert-butoxycarbonyl)amino)-2,5-dimethylhex-2-enoate* (**S4**) as a colorless solid (1.20 g, 88% yield). ^1H NMR (400 MHz, $CDCl_3$) δ 6.54−6.50 (m, 1H), 4.67 (d, J = 8 Hz, 1H), 4.26−4.22 (m, 1H), 3.75 (s, 3H), 1.93 (s, 3H), 1.81−1.73 (m, 1H), 1.43 (s, 9H), 0.96 (d, J = 6.8 Hz, 3H), 0.92 (d, J = 6.8 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 168.5, 155.4, 140.5, 129.5, 79.4, 54.0, 52.0, 33.0, 28.4, 18.6, 18.5, 13.2; HRMS (ESI) m/z: $(M + Na)^+$ Calcd for $C_{14}H_{25}NO_4Na$ 294.1676; Found 294.1683.

To a solution of compound **S4** (1.09 g, 4.0 mmol) in dichloromethane (8.0 mL) was added trifluoroacetic acid (2.0 mL) slowly at 0 °C. The mixture was stirred at room temperature for 30 min. The solvent was then removed in vacuo, and aqueous NaOH (2.0 M, 4.0 mL) was added to the residue. The aqueous layer was extracted with dichloromethane (3 × 5 mL). The combined organic phases were dried over anhydrous sodium sulfate, filtered, and concentrated to give the crude amine as a thick oil, which was used directly in the next step without further purification.

The crude amine was resolved in acetonitrile (4.0 mL), and a 30% aqueous formaldehyde solution (1.0 mL) was added, followed by sodium cyanoborohydride (628 mg, 10 mmol). A few drops of glacial acetic acid were added to maintain the pH near neutrality. The solution was stirred at room temperature for 2 h. After completion of the reaction (detected by TLC), the mixture was concentrated in vacuo. To the residue was added aqueous NaOH (1 M, 5.0 mL), and then it was extracted with ethyl acetate (3 × 5 mL). The combined organic extracts were washed with brine (15 mL), dried over anhydrous sodium sulfate, and concentrated in vacuo to give crude amine **S5**.

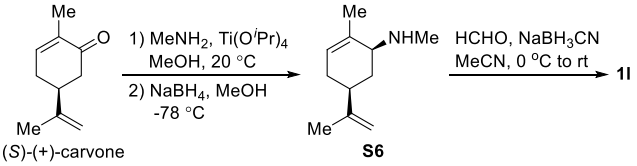
Crude amine **S5** was dissolved in dry tetrahydrofuran (20 mL), and $LiAlH_4$ (159 mg, 4.2 mmol) was added. The mixture was stirred at room temperature for 4 h. Saturated aqueous $KHSO_4$ (5 mL) was added under vigorous stirring. The organic and the water phases were separated. The water phase was extracted with ether (3 × 20 mL). The organic phases were combined, dried over anhydrous sodium sulfate, and evaporated to give the crude alcohol, which was used without further purification.

The crude alcohol and benzyl bromide (684 mg, 4.0 mmol) were stirred vigorously in aqueous NaOH (5.0 mL, 50% NaOH pellets (w/v)) for 6 h. The water phase was extracted with dichloromethane (3 × 10 mL). The combined organic extracts were dried over anhydrous sodium sulfate and concentrated in vacuo. The residue was purified by silica gel column chromatography (petroleum ether:ethyl acetate = 2:1) to give *(S,E)-6-(benzyloxy)-N,N,2,5-tetramethylhex-4-en-3-amine* (**1j**) as a light yellow oil (123 mg, 12% yield (4 steps from **S4**)). $[\alpha]_D^{20}$ = −23.2 (c = 0.41, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.35−7.30 (m, 4H), 7.29−7.27 (m, 1H), 5.37−5.33 (m, 1H), 4.46 (s, 2H), 3.99 (s, 2H), 2.73 (dd, J = 10.4, 7.6 Hz, 1H), 2.22 (s, 6H), 1.89−1.80 (m, 1H), 1.71 (d, J = 1.2 Hz, 3H), 0.95 (d, J = 6.8 Hz, 3H), 0.84 (d, J = 6.8 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 138.6, 136.4, 128.5, 127.8, 127.7, 125.6, 76.3, 71.3, 68.0, 42.1, 29.9, 20.4, 18.5, 14.8;

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HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{17}H_{28}NO$ 262.2165; Found 262.2173.

Preparation of Allylamine 1l.[19]



A mixture of (*S*)-(+)-carvone (300 mg, 2.0 mmol), titanium(IV) isopropoxide (1.2 mL, 1.14 g, 4.0 mmol), and methylamine (2.0 M in methanol, 5.0 mL, 10 mmol) in absolute methanol (10 mL) was stirred under nitrogen at room temperature for 12 h. Sodium borohydride (113 mg, 3.0 mmol) was then added at −78 °C, and the resulting mixture was stirred for an additional 2 h. The reaction was then quenched by adding water (3 mL), and stirring was maintained at room temperature for 20 min. The resulting inorganic precipitate was filtered off over a pad of Celite and washed with methanol and ethyl acetate. The combined organic extracts were dried over anhydrous sodium sulfate. The organic layer was concentrated under reduced pressure to give crude compound **S6**, which was used directly for the next step.

Crude compound **S6** was resolved in acetonitrile (3.0 mL), and a 30% aqueous formaldehyde solution (0.40 mL) was added, followed by sodium cyanoborohydride (81.7 mg, 1.3 mmol). A few drops of glacial acetic acid were added to maintain the pH near neutrality. The mixture was stirred at room temperature for 2 h. After completion of the reaction (detected by TLC), the mixture was concentrated in vacuo. To the residue was added aqueous NaOH (2 N, 10 mL), and the mixture was extracted with ethyl acetate (3 × 15 mL). Combined organic extracts were washed with brine (20 mL), dried over anhydrous sodium sulfate, and concentrated in vacuo. The residue was purified by column chromatography (dichloromethane/methanol = 15:1) to give *(1S,5S)-N,N,2-trimethyl-5-(prop-1-en-2-yl)cyclohex-2-en-1-amine* (**1l**) as a light yellow oil (115 mg, 64% yield). $[\alpha]_D^{20}$ = −15.9 (c = 0.27, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.67−5.64 (m, 1H), 4.77−4.65 (m, 2H), 3.02−2.99 (m, 1H), 2.29 (s, 6H), 2.27−2.21 (m, 1H), 2.10−2.02 (m, 2H), 1.91−1.79 (m, 1H), 1.75−1.73 (m, 6H), 1.35−1.27 (m, 1H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 149.6, 134.4, 125.7, 108.8, 61.6, 41.4, 38.5, 30.6, 26.1, 22.0, 21.1; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{12}H_{22}N$ 180.1747; Found 180.1750. The dr value was determined to be 95:5 by ^{1}H NMR spectroscopic analysis.

General Procedure for the Screening of Solvents (Table 1). To a stirred solution of *(E)-1-(4-phenylbut-3-en-2-yl)pyrrolidine* (***rac*-1a**, 40.3 mg, 0.20 mmol) in a solvent (1.0 mL) was added *1-phenylbuta-2,3-dien-1-one* (**2a**, 34.6 mg, 0.24 mmol). The mixture was stirred at room temperature for 12 h and then concentrated under reduced pressure to give crude product *(2Z,6E)-1,5-diphenyl-3-(pyrrolidin-1-yl)octa-2,6-dien-1-one* (***rac*-3a**) as an orange oil (NMR yields are given in Table 1). ^{1}H NMR (500 MHz, $CDCl_3$) δ 7.89−7.87 (m, 2H), 7.41−7.36 (m, 3H), 7.33−7.31 (m, 2H), 7.28−7.25 (m, 2H), 7.19−7.15 (m, 1H), 5.86−5.81 (m, 1H), 5.64−5.57 (m, 3H), 3.91−3.85 (m, 1H), 3.82−3.76 (m, 1H), 3.32−3.18 (m, 3H), 3.06−2.96 (m, 1H), 2.85−2.79 (m, 1H), 1.85−1.72 (m, 3H), 1.70 (d, J = 6.5 Hz, 3H), 1.60−1.54 (m, 1H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 186.8, 163.9, 144.4, 143.5, 133.1, 130.0, 128.3, 128.0, 127.3, 126.4, 125.5, 92.8, 48.8, 48.4, 47.6, 38.1, 25.3, 24.8, 18.2; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{24}H_{28}NO$ 346.2165; Found 346.2175.

General Procedure for the Aza-Claisen Rearrangement between Allylamines and Allenones (Tables 2−4). To a stirred solution of allylamine **1** (0.20 mmol) in methanol (1.0 mL) was added allenone **2** (0.24 mmol). The mixture was stirred at room temperature for 12 h and concentrated under reduced pressure to remove the methanol. The residue was diluted with dichloromethane (5.0 mL, containing 1.0% (v/v) water), and silica gel (500 mg) was added. The mixture was stirred at room temperature for 12 h and filtered, and the silica gel was washed with dichloromethane (3 × 10 mL). The combined filtrates were concentrated under reduced pressure, and the residue was purified by silica gel chromatography (petroleum ether:ethyl acetate = 10:1) to give compound **4**.

The absolute configuration of product **4o** was assigned to be *S* by X-ray crystallography (CCDC 1489729), and that of the rest of the products (shown in Tables 3 and 4) was assigned by analogy (see below). All of products **4** exist predominantly in the enol form according to the NMR spectra.

(S,E)-1,5-Diphenyloct-6-ene-1,3-dione (***4a***). Yellow solid (55.6 mg, 95% yield); mp 43−44 °C; $[\alpha]_D^{20}$ = −15.9 (c = 0.27, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.82−7.80 (m, 2H), 7.53−7.50 (m, 1H), 7.45−7.41 (m, 2H), 7.33−7.29 (m, 2H), 7.24−7.18 (m, 3H), 6.06 (s, 1H), 5.66 (dd, J = 7.2, 1.6 Hz, 1H), 5.63−5.46 (m, 1H), 3.90−3.85 (m, 1H), 2.81 (d, J = 7.6 Hz, 2H), 1.66 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.8, 183.4, 143.7, 135.1, 133.4, 132.4, 128.7, 128.7, 127.6, 127.1, 126.6, 126.0, 97.3, 45.8, 45.4, 18.1; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{20}H_{21}O_2$ 293.1536; Found 293.1535. The ee was determined to be 97% by HPLC analysis (Chiralpak IC column, λ = 254 nm, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min): t_R (minor) = 22.39 min, t_R (major) = 28.37 min.

(S,E)-1,5-Diphenylnon-6-ene-1,3-dione (***4b***). Yellow solid (50.9 mg, 83% yield); mp 51−52 °C; $[\alpha]_D^{20}$ = −19.3 (c = 0.45, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.81 (d, J = 7.6 Hz, 2H), 7.48 (d, J = 7.2 Hz, 1H), 7.42−7.40 (m, 2H), 7.32−7.28 (m, 2H), 7.24−7.19 (m, 3H), 6.05 (s, 1H), 5.64−5.49 (m, 2H), 3.90−3.83 (m, 1H), 2.80 (d, J = 7.6 Hz, 2H), 2.04−1.99 (m, 2H), 0.93 (t, J = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.7, 183.4, 143.8, 135.1, 133.0, 132.4, 131.1, 128.8, 128.7, 127.7, 127.6, 127.1, 126.6, 97.3, 45.9, 45.4, 25.7, 13.8; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{21}H_{23}O_2$ 307.1693; Found 307.1695. The ee was determined to be 97% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, n-hexane/isopropanol = 99:1, flow rate = 0.8 mL/min): t_R (minor) = 9.12 min, t_R (major) = 10.22 min.

(S,E)-1,5,9-Triphenylnon-6-ene-1,3-dione (***4c***). Yellow oil (68.8 mg, 90% yield); $[\alpha]_D^{20}$ = −17.5 (c = 1.85, $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$) δ 16.12 (s, 1H), 7.82−7.80 (m, 2H), 7.53−7.50 (m, 1H), 7.45−7.41 (m, 2H), 7.31−7.09 (m, 10H), 6.04 (s, 1H), 5.62 (dd, J = 15.5, 7.5 Hz, 1H), 5.55−5.48 (m, 1H), 3.90−3.83 (m, 1H), 2.78 (d, J = 7.5 Hz, 2H), 2.63 (t, J = 7.5 Hz, 2H), 2.31 (q, J = 7.5 Hz, 2H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 194.8, 183.3, 143.6, 141.9, 135.0, 132.9, 132.4, 130.4, 128.7, 128.6, 128.4, 127.6, 127.1, 126.6, 125.9, 97.3, 45.7, 45.3, 35.9, 34.5; HRMS (ESI) m/z: $(M + Na)^+$ Calcd for $C_{27}H_{26}O_2Na$ 405.1825; Found 405.1838.The ee was determined to be 96% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, n-hexane/isopropanol = 97:3, flow rate = 1.0 mL/min): t_R (minor) = 9.65 min, t_R (major) = 12.62 min.

(S,E)-8-Methyl-1,5-diphenylnon-6-ene-1,3-dione (***4d***). Yellow solid (57.7 mg, 90% yield); mp 61−62 °C; $[\alpha]_D^{20}$ = −14.6 (c = 0.48, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.82−7.79 (m, 2H), 7.52−7.48 (m, 1H), 7.44−7.40 (m, 2H), 7.32−7.28 (m, 2H), 7.24−7.18 (m, 3H), 6.05 (s, 1H), 5.56 (dd, J = 15.6, 7.2 Hz, 1H), 5.48 (dd, J = 15.6, 7.2 Hz, 1H), 3.88−3.81 (m, 1H), 2.79 (d, J = 8.0 Hz, 2H), 2.28−2.20 (m, 1H), 0.94 (d, J = 6.4 Hz, 6H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.7, 183.5, 143.8, 138.6, 135.1, 132.4, 128.9, 128.8, 128.7, 127.6, 127.1, 126.6, 97.4, 46.0, 45.4, 31.1, 22.6; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{22}H_{25}O_2$ 321.1849; Found 321.1853. The ee was determined to be 98% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, n-hexane/isopropanol = 99:1, flow rate = 0.8 mL/min): t_R (minor) = 10.43 min, t_R (major) = 11.92 min.

(S,E)-1,5,7-Triphenylhept-6-ene-1,3-dione (***4e***). Brown solid (55.3 mg, 78% yield); mp 112−113 °C; $[\alpha]_D^{20}$ = −15.9 (c = 0.27, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ 16.09 (s, 1H), 7.79 (d, J = 7.6 Hz, 2H), 7.51−7.17 (m, 13H), 6.46−6.31 (m, 2H), 6.09 (s, 1H), 4.14−4.09 (m, 1H), 2.99−2.89 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.4, 183.4, 142.9, 137.2, 134.9, 132.4, 130.4, 128.8, 128.7, 128.6, 127.8, 127.5, 127.1, 126.9, 126.4, 97.3, 45.6, 45.5; HRMS (ESI) m/z: $(M + Na)^+$ Calcd for $C_{25}H_{22}O_2Na$ 377.1512; Found 377.1519. The ee was determined to be 96% by HPLC analysis (Chiralpak OJ

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column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 10.68 min, t_R (major) = 13.12 min.

(S,E)-5-(2-Chlorophenyl)-1-phenyloct-6-ene-1,3-dione (**4f**). Yellow oil (58.8 mg, 90% yield); $[\alpha]_D^{20}$ = −17.5 (*c* = 1.85, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 16.05 (s, 1H), 7.84 (d, *J* = 7.2 Hz, 2H), 7.53−7.49 (m, 1H), 7.45−7.44 (m, 2H), 7.42−7.33 (m, 1H), 7.28−7.22 (m, 2H), 7.20−7.12 (m, 1H), 6.14 (s, 1H), 5.65−5.51 (m, 2H), 4.44−4.40 (m, 1H), 2.89−2.76 (m, 2H), 1.67 (d, *J* = 5.6 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.8, 183.1, 141.0, 135.0, 133.7, 132.4, 131.5, 130.0, 128.7, 128.5, 127.8, 127.2, 127.1, 127.0, 96.9, 44.6, 41.3, 18.2; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{20}H_{19}O_2ClNa$ 349.0966; Found 349.0964. The ee was determined to be 99% by HPLC analysis (Chiralpak OJ column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 11.55 min, t_R (major) = 18.90 min.

(S,E)-5-(4-Methoxyphenyl)-1-phenylnon-6-ene-1,3-dione (**4g**). Yellow solid (60.5 mg, 90% yield); mp 72−73 °C; $[\alpha]_D^{20}$ = −10.6 (*c* = 0.64, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 16.12 (s, 1H), 7.82 (d, *J* = 6.8 Hz, 2H), 7.52−7.41 (m, 3H), 7.16−7.14 (m, 2H), 6.85−6.83 (m, 2H), 6.05 (s, 1H), 5.61−5.47 (m, 2H), 3.87−3.80 (m 1H), 3.77 (s, 3H), 2.77 (dd, *J* = 7.6, 2.4 Hz, 2H), 2.04−1.92 (m, 2H), 0.93 (t, *J* = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.9, 183.4, 158.2, 135.8, 135.1, 132.7, 132.4, 131.4, 128.7, 128.5, 127.1, 114.0, 97.3, 55.3, 46.0, 44.5, 25.7, 13.9; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{22}H_{24}O_3Na$ 359.1618; Found 359.1628. The ee was determined to be 80% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, *n*-hexane/isopropanol = 97:3, flow rate = 1.0 mL/min): t_R (minor) = 6.93 min, t_R (major) = 7.88 min.

(S,E)-1-Phenyl-5-((E)-styryl)oct-6-ene-1,3-dione (**4h**). Orange oil (56.7 mg, 89% yield); $[\alpha]_D^{20}$ = −11.7 (*c* = 1.2, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.87 (d, *J* = 7.2 Hz, 2H), 7.54−7.50 (m, 1H), 7.46−7.42 (m, 2H), 7.36−7.30 (m, 2H), 7.28−7.26 (m, 2H), 7.21−7.20 (m, 1H), 6.45 (d, *J* = 16.0 Hz, 1H), 6.20−6.14 (d, *J* = 16.0 Hz, 1H), 6.18 (s, 1H), 5.61−5.45 (m, 2H), 3.49−3.42 (m, 1H), 2.60 (d, *J* = 7.2 Hz, 2H), 1.69 (d, *J* = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.7, 183.7, 137.4, 135.1, 132.4, 132.1, 132.0, 130.2, 130.1, 128.9, 128.7, 128.6, 128.4, 127.3, 127.2, 126.5, 126.3, 97.3, 44.9, 42.9, 18.2; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{22}H_{22}O_2Na$ 341.1512; Found 341.1518. The ee was determined to be 99% by HPLC analysis (Chiralpak OJ column, λ = 254 nm, *n*-hexane/isopropanol = 90:10, flow rate = 1.0 mL/min): t_R (minor) = 6.41 min, t_R (major) = 15.82 min.

(R,E)-5-Cyclohexyl-1-phenyloct-6-ene-1,3-dione (**4i**). Yellow oil (52.5 mg, 88% yield); $[\alpha]_D^{20}$ = −17.5 (*c* = 1.85, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 7.93−7.86 (m, 2H), 7.57−7.56 (m, 1H), 7.52−7.42 (m, 2H), 6.12 (s, 1H), 5.43−5.34 (m, 1H), 5.29−5.23 (m, 1H), 2.55 (dd, *J* = 13.5, 5.0 Hz, 1H), 2.41−2.29 (m, 2H), 1.74−1.60 (m, 7H), 1.32−0.86 (m, 7H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 196.7, 183.2, 135.3, 132.3, 132.0, 128.7, 127.1, 126.5, 97.2, 46.0, 42.6, 42.1, 31.2, 29.5, 26.7, 18.1; HRMS (ESI) *m*/*z*: $(M + H)^+$ Calcd for $C_{20}H_{27}O_2$ 299.2006; Found 299.2012. The ee was determined to be 97% by HPLC analysis (Performed on Thermo UltiMate 3000, Chiralpak AD-H column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 10.50 min, t_R (major) = 11.53 min.

(S,E)-5-((Benzyloxy)methyl)-5,8-dimethyl-1-phenylnon-6-ene-1,3-dione (**4j**). Yellow oil (65.8 mg, 87% yield); $[\alpha]_D^{20}$ = −18.1 (*c* = 0.75, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.84 (d, *J* = 7.2 Hz, 2H), 7.51 (t, *J* = 7.2 Hz, 1H), 7.42 (t, *J* = 7.6 Hz, 2H), 7.34−7.27 (m, 5H), 6.15 (s, 1H), 5.50 (dd, *J* = 16.0, 1.2 Hz, 1H), 5.40 (dd, *J* = 16.0, 6.8 Hz, 1H), 4.54 (s, 2H), 3.35 (d, *J* = 8.8 Hz, 1H), 3.30 (d, *J* = 8.8 Hz, 1H), 2.52 (s, 2H), 2.29−2.24 (m, 1H), 1.17 (s, 3H), 0.95 (d, *J* = 6.8 Hz, 3H), 0.94 (d, *J* = 6.8 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.2, 184.3, 138.8, 136.2, 135.5, 132.3, 132.1, 128.7, 128.4, 127.6, 127.5, 127.1, 98.5, 77.8, 73.3, 47.1, 40.8, 31.4, 22.8, 22.4; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{25}H_{30}O_3Na$ 401.2087; Found 401.2083. The ee was determined to be 95% by HPLC analysis (Chiralpak OJ-H column, λ = 254 nm, *n*-hexane/isopropanol = 98:2, flow rate = 1.0 mL/min): t_R (minor) = 10.53 min, t_R (major) = 11.80 min.

(S,E)-6-Methyl-1,5-diphenyloct-6-ene-1,3-dione (**4k**). Yellow solid (52.0 mg, 85% yield); mp 63−65 °C; $[\alpha]_D^{20}$ = −67.4 (*c* = 0.53, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 16.08 (s, 1H), 7.80 (d, *J* = 7.2 Hz, 2H), 7.52−7.50 (m, 1H), 7.49−7.48 (m, 2H), 7.30−7.16 (m, 5H), 6.05 (s, 1H), 5.51−5.45 (m, 1H), 3.86 (t, *J* = 7.6 Hz, 1H), 2.95 (dd, *J* = 10.8, 7.6 Hz, 1H), 2.83 (dd, *J* = 10.8, 7.6 Hz, 1H), 1.62 (d, *J* = 6.8 Hz, 3H), 1.51 (s, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 195.8, 182.9, 142.9, 137.0, 135.0, 132.3, 128.7, 128.5, 127.9, 127.1, 126.6, 119.6, 97.1, 50.3, 43.4, 14.8, 13.6; HRMS (ESI) *m*/*z*: $(M + H)^+$ Calcd for $C_{21}H_{23}O_2$ 307.1693; Found 307.1698. The ee was determined to be 99% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 6.00 min, t_R (major) = 7.49 min.

4-((1S,5R)-2-Methyl-5-(prop-1-en-2-yl)cyclohex-2-en-1-yl)-1-phenylbutane-1,3-dione (**4l**). Yellow oil (53.9 mg, 91% yield); $[\alpha]_D^{20}$ = −16.1 (*c* = 0.49, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.90 (d, *J* = 7.6 Hz, 2H), 7.53 (t, *J* = 7.2 Hz, 1H), 7.51−7.44 (m, 2H), 6.19 (s, 1H), 5.51−5.48 (m, 1H), 4.72 (s, 2H), 2.73 (dd, *J* = 14.4, 3.6 Hz, 1H), 2.63−2.61 (m, 1H), 2.39−2.33 (m, 1H), 2.13−2.08 (m, 1H), 1.95−1.88 (m, 1H), 1.73 (s, 6H), 1.77−1.68 (m, 1H),1.63−1.54 (m, 2H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 196.0, 183.7, 149.6, 135.5, 135.1, 132.5, 128.8, 127.2, 123.1, 109.0, 97.2, 42.8, 36.8, 36.0, 32.0, 31.0, 22.1, 21.0; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{20}H_{24}O_2Na$ 319.1669; Found 319.1668. The dr value was determined to be 93:7 by ^1H NMR spectroscopic analysis.

(S,E)-1-(4-Methoxyphenyl)-5-phenyloct-6-ene-1,3-dione (**4m**). Yellow solid (56.7 mg, 88% yield), mp 72−73 °C; $[\alpha]_D^{20}$ = −28.1 (*c* = 0.59, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.80−7.78 (m, 2H), 7.31−7.20 (m, 5H), 6.90 (d, *J* = 8.8 Hz, 2H), 5.99 (s, 1H), 5.64 (dd, *J* = 14.8, 6.8 Hz, 1H), 5.53−5.44 (m, 1H), 3.89−3.83 (m, 1H), 3.83 (s, 3H), 2.76 (d, *J* = 7.6 Hz, 2H), 1.64 (d, *J* = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 192.6, 184.1, 163.1, 143.8, 133.4, 131.2, 129.2, 128.6, 127.6, 127.6, 127.6, 126.5, 125.8, 114.0, 96.4, 55.5, 45.4, 45.3, 18.1; HRMS (ESI) *m*/*z*: $(M + H)^+$ Calcd for $C_{21}H_{23}O_3$ 323.1642; Found 323.1646. The ee was determined to be 97% by HPLC analysis (Chiralpak IC column, λ = 254 nm, *n*-hexane/isopropanol = 98:2, flow rate = 1.0 mL/min): t_R (major) = 14.58 min, t_R (minor) = 16.63 min.

(S,E)-1-(4-Fluorophenyl)-5-phenyloct-6-ene-1,3-dione (**4n**). Brown solid (49.0 mg, 79% yield), mp 54−56 °C; $[\alpha]_D^{20}$ = −19.2 (*c* = 0.41, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.84−7.80 (m, 2H), 7.32−7.18 (m, 7H), 5.99 (s, 1H), 5.64 (dd, *J* = 14.8, 7.2 Hz, 1H), 5.54−5.46 (m, 1H), 3.87−3.83 (m, 1H), 2.80 (d, *J* = 7.6 Hz, 2H), 1.66 (d, *J* = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 193.9, 183.0, 166.7 (C-F, $^1J_{C\text{-}F}$ = 252.0 Hz), 164.2 (C-F, $^1J_{C\text{-}F}$ = 252.0 Hz), 143.7, 133.3, 129.6 (C-F, $^3J_{C\text{-}F}$ = 9.0 Hz), 129.5 (C-F, $^3J_{C\text{-}F}$ = 9.0 Hz), 128.7, 127.6, 126.7, 126.0, 116.0 (C-F, $^2J_{C\text{-}F}$ = 11.7 Hz), 115.7 (C-F, $^2J_{C\text{-}F}$ = 11.7 Hz), 97.0, 45.6, 45.5, 18.1; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{20}H_{19}O_2FNa$ 333.1261; Found 333.1269. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 10.85 min, t_R (major) = 12.24 min.

(S,E)-1-(4-Bromophenyl)-5-phenyloct-6-ene-1,3-dione (**4o**). Yellow solid (66.1 mg, 89% yield); mp 100−102 °C; $[\alpha]_D^{20}$ = −18.1 (*c* = 0.35, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 16.02 (s, 1H), 8.00 (d, *J* = 8.0 Hz, 2H), 7.57 (d, *J* = 8.0 Hz, 2H), 7.32−7.18 (m, 5H), 6.00 (s, 1H), 5.64 (dd, *J* = 15.6, 7.2 Hz, 1H), 5.56−5.45 (m, 1H), 3.89−3.83 (m, 1H), 2.80 (d, *J* = 7.2 Hz, 2H), 1.65 (d, *J* = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 195.0, 182.3, 143.6, 134.0, 133.3, 132.0, 128.7, 128.6, 127.6, 127.2, 126.7, 126.0, 97.2, 45.8, 45.4, 18.1; HRMS (ESI) *m*/*z*: $(M + Na)^+$ Calcd for $C_{20}H_{19}O_2NaBr$ 393.0461; Found 393.0465. The ee was determined to be 97% by HPLC analysis (Chiralpak OD-H column, λ = 254 nm, *n*-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 8.69 min, t_R (major) = 15.44 min.

(S,E)-1-(2-Chlorophenyl)-5-phenyloct-6-ene-1,3-dione (**4p**). Brown oil (52.3 mg, 80% yield); $[\alpha]_D^{20}$ = −13.1 (*c* = 0.29, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 15.60 (s, 1H), 7.50 (dd, *J* = 7.6, 2.0 Hz, 1H), 7.42−7.18 (m, 8H), 5.93 (s, 1H), 5.67−5.60 (m, 1H), 5.55−5.47 (m, 1H), 3.85 (q, *J* = 7.6 Hz, 1H), 2.80−2.76 (m, 2H),

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1.66 (d, J = 6.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 193.6, 184.7, 143.5, 135.7, 133.2, 131.8, 131.7, 130.7, 130.1, 128.7, 127.6, 127.0, 126.7, 126.0, 102.6, 45.5, 45.4, 18.1; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{20}H_{20}O_2Cl$ 327.1146; Found 327.1146. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 254 nm, n-hexane/isopropanol = 98:2, flow rate = 1.0 mL/min): t_R (minor) = 8.08 min, t_R (major) = 8.75 min.

*(S,E)-1-Mesityl-5-phenyloct-6-ene-1,3-dione (**4q**).* Yellow oil (40.8 mg, 61% yield); $[\alpha]_D^{20}$ = −26.1 (c = 0.39, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 15.66 (s, 1H), 7.30−7.27 (m, 2H), 7.24−7.19 (m, 3H), 6.83 (s, 2H), 5.64−5.60 (m, 1H), 5.55−5.46 (m, 1H), 5.48 (s, 1H), 3.85−3.79 (m, 1H), 2.77−2.66 (m, 2H), 2.27 (s, 3H), 2.14 (s, 6H), 1.66 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 194.6, 188.9, 143.5, 138.9, 134.7, 134.4, 133.3, 128.7, 128.4, 127.5, 126.6, 125.8, 103.7, 45.8, 45.7, 21.2, 19.5, 18.1; HRMS (ESI) m/z: $(M + Na)^+$ Calcd for $C_{23}H_{26}O_2Na$ 357.1825; Found 357.1839. The ee was determined to be 97% by HPLC analysis (Chiralpak IC column, λ = 254 nm, n-hexane/isopropanol = 97:3, flow rate = 1.0 mL/min): t_R (minor) = 7.12 min, t_R (major) = 8.45 min.

*(S,E)-1-(Furan-2-yl)-5-phenyloct-6-ene-1,3-dione (**4r**).* Yellow oil (46.9 mg, 83% yield); $[\alpha]_D^{20}$ = −25.9 (c = 1.21, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 15.48 (s, 1H), 7.54 (s, 1H), 7.31−7.17 (m, 5H), 7.10 (d, J = 3.5 Hz, 1H), 6.51 (dd, J = 3.5, 2.0 Hz, 1H), 5.98 (s, 1H), 5.64−5.58 (m, 1H), 5.52−5.41 (m, 1H), 3.89−3.81 (m, 1H), 2.74 (d, J = 7.5 Hz, 2H), 1.65 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 190.7, 176.0, 150.6, 146.1, 143.6, 133.3, 128.7, 127.5, 126.6, 126.0, 115.7, 112.6, 96.7, 45.4, 44.6, 18.0; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{18}H_{19}O_3$ 283.1329; Found 283.1334. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 254 nm, n-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (major) = 9.73 min, t_R (minor) = 11.42 min.

*(S,E)-5-Phenyl-1-(thiophen-2-yl)oct-6-ene-1,3-dione (**4s**).* Yellow oil (49.5 mg, 83% yield); $[\alpha]_D^{20}$ = −18.9 (c = 0.83, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 7.61−7.60 (m, 1H), 7.57−7.56 (m, 1H), 7.32−7.28 (m, 2H), 7.25−7.21 (m, 3H), 7.11−7.07 (m, 1H), 5.89 (s, 1H), 5.66−5.60 (m, 1H), 5.54−5.45 (m, 1H), 3.82 (q, J = 7.6 Hz, 1H), 2.74 (d, J = 7.6 Hz, 2H), 1.65 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 188.3, 181.8, 143.6, 141.8, 133.2, 132.5, 130.3, 128.7, 128.3, 127.5, 126.6, 126.1, 97.1, 45.6, 44.1, 18.1; HRMS (ESI) m/z: $(M + Na)^+$ Calcd for $C_{18}H_{18}O_2SNa$ 321.0920; Found 321.0923. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 220 nm, n-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 9.33 min, t_R (major) = 12.20 min.

*(S,1E,8E)-1,7-Diphenyldeca-1,8-diene-3,5-dione (**4t**).* Yellow oil (47.8 mg, 75% yield); $[\alpha]_D^{20}$ = −22.9 (c = 0.92, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$) δ 15.31 (s, 1H), 7.55 (d, J = 16.0 Hz, 1H), 7.51−7.48 (m, 2H), 7.39−7.34 (m, 3H), 7.32−7.28 (m, 2H), 7.24−7.17 (m, 3H), 6.42 (d, J = 16.0 Hz, 1H), 5.61 (dd, J = 16.0, 7.2 Hz, 1H), 5.56 (s, 1H), 5.52−5.43 (m, 1H), 3.87 (q, J = 7.6 Hz, 1H), 2.78 (d, J = 7.2 Hz, 2H), 1.65 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 199.1, 176.8, 143.8, 139.8, 135.2, 133.4, 130.0, 129.0, 128.7, 128.0, 127.6, 126.6, 125.8, 122.9, 101.7, 46.8, 45.1, 18.1; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{22}H_{23}O_2$ 319.1693; Found 319.1691. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 220 nm, n-hexane/isopropanol = 90:10, flow rate = 1.0 mL/min): t_R (major) = 29.02 min, t_R (minor) = 32.16 min.

*(S,E)-1,6-Diphenylnon-7-ene-2,4-dione (**4u**).* Yellow oil (50.2 mg, 82% yield); $[\alpha]_D^{20}$ = −28.8 (c = 1.12, $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$) δ 15.33 (s, 1H), 7.33−7.24 (m, 5H), 7.20−7.13 (m, 5H), 5.55 (dd, J = 15.5, 7.5 Hz, 1H), 5.45−5.39 (m, 1H), 5.32 (s, 1H), 3.80−3.67 (m, 1H), 3.52 (s, 2H), 2.61 (d, J = 7.5 Hz, 2H), 1.62 (d, J = 6.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$) δ 192.4, 192.2, 143.6, 135.2, 133.3, 129.4, 128.8, 128.6, 127.5, 127.1, 126.6, 125.8, 100.6, 45.3, 45.2, 44.8, 18.0; HRMS (ESI) m/z: $(M + H)^+$ Calcd for $C_{21}H_{23}O_2$ 307.1693; Found 307.1698. The ee was determined to be 97% by HPLC analysis (Chiralpak OJ-H column, λ = 254 nm, n-hexane/isopropanol = 95:5, flow rate = 1.0 mL/min): t_R (minor) = 9.28 min, t_R (major) = 9.94 min.

A 5 mmol Scale Reaction. To a stirred solution of allylamine **1a** (1.01 g, 5.0 mmol) in methanol (25 mL) was added allenone **2a** (865 mg, 6.0 mmol). The mixture was stirred at room temperature for 12 h and concentrated under reduced pressure to remove the methanol. The residue was diluted with dichloromethane (50 mL, containing 1.0% (v/v) water), and silica gel (12.5 g) was added. The mixture was stirred at room temperature for 12 h and then filtered, and the silica gel was washed with dichloromethane (3 × 25 mL). The combined filtrates were concentrated under reduced pressure, and the residue was purified by silica gel chromatography (petroleum ether:ethyl acetate = 10:1) to give compound **4a** as a yellow solid (1.27 g, 87%).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.0c02721.

X-ray crystallographic data and copies of NMR spectra and LC traces (PDF)

Accession Codes

CCDC 1489729 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) For reviews, see: (a) Enders, D.; Knopp, M.; Schiffers, R. Asymmetric [3.3]-Sigmatropic Rearrangements in Organic Synthesis. *Tetrahedron: Asymmetry* **1996**, *7*, 1847−1882. (b) Martin Castro, A. M. Claisen Rearrangement over the Past Nine Decades. *Chem. Rev.* **2004**, *104*, 2939−3002. (c) Nubbemeyer, U. Recent Advances in Charge-Accelerated Aza-Claisen Rearrangements. *Top. Curr. Chem.* **2005**, *244*, 149−213. (d) Majumdar, K. C.; Bhattacharyya, T.; Chattopadhyay, B.; Sinha, B. Recent Advances in the Aza-Claisen Rearrangement. *Synthesis* **2009**, *2009*, 2117−2142. (e) Jung, J.-W.; Kim, S.-H.; Suh, Y.-G. Advances in Aza-Claisen-Rearrangement-Induced Ring-Expansion Strategies. *Asian J. Org. Chem.* **2017**, *6*, 1117−1129.

(2) (a) Kandeel, K. A.; Vernon, J. M. Reactions of Tertiary Allylamines with Dimethyl Acetylenedicarboxylate. *J. Chem. Soc., Perkin Trans. 1* **1987**, 2023−2026. (b) Schwan, A. L.; Warkentin, J. The reactions of simple dimethylallylamines with dimethyl acetylenedicarboxylate. Formation of 1-dimethylamino-2-allylmaleates via formal allyl transfer. *Can. J. Chem.* **1988**, *66*, 1686−1694. (c) Vedejs, E.; Gingras, M. Aza-Claisen Rearrangements Initiated by Acid-Catalyzed Michael Addition. *J. Am. Chem. Soc.* **1994**, *116*, 579−588.

(3) (a) Weston, M. H.; Nakajima, K.; Parvez, M.; Back, T. G. Ring-expansion of tertiary cyclic a-vinylamines by tandem conjugate addition to (p-toluenesulfonyl)ethyne and formal 3-aza-Cope rearrangement. *Chem. Commun.* **2006**, 3903−3905. (b) Weston, M. H.; Nakajima, K.; Back, T. G. Tandem Conjugate Additions and 3-Aza-Cope Rearrangements of Tertiary Allyl Amines and Cyclic α-Vinylamines with Acetylenic Sulfones. Applications to Simple and Iterative Ring Expansions Leading to Medium and Large-Ring Nitrogen Heterocycles. *J. Org. Chem.* **2008**, *73*, 4630−4637.

(4) (a) Lambert, T. H.; MacMillan, D. W. C. Development of a New Lewis Acid-Catalyzed [3,3]-Sigmatropic Rearrangement: The Allenoate-Claisen Rearrangement. *J. Am. Chem. Soc.* **2002**, *124*, 13646−13647. (b) Lopes, S. M. M.; Santos, B. S.; Palacios, F.; Pinho e Melo, T. M. V. D. Microwave-assisted reactions of allenic esters: [3 + 2] anellations and allenoate-Claisen rearrangement. *ARKIVOC* **2010**, *2010*, 70−81. (c) Miró, J.; Gensch, T.; Ellwart, M.; Han, S.-J.; Lin, H.-H.; Sigman, M. S.; Toste, F. D. Enantioselective Allenoate-Claisen Rearrangement Using Chiral Phosphate Catalysts. *J. Am. Chem. Soc.* **2020**, *142*, 6390−6399.

(5) For selected examples: (a) Roberts, S. M.; Smith, C.; Thomas, R. J. Some Reactions of 2-Azabicyclo[2.2.1] hept-5-enes with Diphenylketene: Preparation of Polysubstituted Piperidin-2-ones. *J. Chem. Soc., Perkin Trans. 1* **1990**, 1493−1495. (b) Cid, M. M.; Eggnauer, U.; Weber, H. P.; Pombo-Villar, E. Synthesis of (−)-δ-N-Normethylskytanthine. *Tetrahedron Lett.* **1991**, *32*, 7233−7236. (c) Edstrom, E. D. New Methodology for the Synthesis of Functionalized Indolizidine and Quinolizidine Ring Systems. *J. Am. Chem. Soc.* **1991**, *113*, 6690−6692. (d) Diederich, M.; Nubbemeyer, U. Synthesis of Optically Active Nine-Membered Ring Lactams by a Zwitterionic Aza-Claisen Reaction. *Angew. Chem., Int. Ed. Engl.* **1995**, *34*, 1026−1028. (e) Diederich, M.; Nubbemeyer, U. Diastereoselective Zwitterionic Aza-Claisen Rearrangement: Synthesis of Nine-Membered Ring Lactams and Transannular Ring Contraction. *Chem. - Eur. J.* **1996**, *2*, 894−900. (f) Sudau, A.; Nubbemeyer, U. Unusual Diastereoselection in the Synthesis of Nine-Membered Ring Lactams and Conformation-Controlled Transannular Reactions to Generate Optically Active Indolizidinones. *Angew. Chem., Int. Ed.* **1998**, *37*, 1140−1143. (g) Laabs, S.; Scherrmann, A.; Sudau, A.; Diederich, M.; Kierig, C.; Nubbemeyer, U. Zwitterionic Aza-Claisen Rearrangement: Reaction of Carboxylic Acid Fluorides with N-Allyl Amines. *Synlett* **1999**, *1999*, 25−28. (h) Sudau, A.; Münch, W.; Nubbemeyer, U.; Bats, J. W. Planar Chirality: Synthesis and Transannular Reactions of Unsaturated Optically Active Azoninones Bearing E-Olefins. *J. Org. Chem.* **2000**, *65*, 1710−1720. (i) Sudau, A.; Münch, W.; Bats, J. W.; Nubbemeyer, U. Synthesis of the Bicyclic Core of Pumiliotoxins. *Eur. J. Org. Chem.* **2002**, *2002*, 3304−3314.

(6) (a) Yoon, T. P.; Dong, V. M.; MacMillan, D. W. C. Development of a New Lewis Acid-Catalyzed Claisen Rearrangement. *J. Am. Chem. Soc.* **1999**, *121*, 9726−9727. (b) Yoon, T. P.; MacMillan, D. W. C. Enantioselective Claisen Rearrangements: Development of a First Generation Asymmetric Acyl-Claisen Reaction. *J. Am. Chem. Soc.* **2001**, *123*, 2911−2912. (c) Dong, V. M.; MacMillan, D. W. C. Design of a New Cascade Reaction for the Construction of Complex Acyclic Architecture: The Tandem Acyl-Claisen Rearrangement. *J. Am. Chem. Soc.* **2001**, *123*, 2448−2449. (d) Tenza, K.; Northen, J. S.; O'Hagan, D.; Slawin, A. M. Z. Stereoselective α-fluoroamide and α-fluoro-γ-lactone synthesis by an asymmetric zwitterionic aza-Claisen rearrangement. *Beilstein J. Org. Chem.* **2005**, *1*, 13. (e) Craig, D.; King, N. P.; Mountford, D. M. Silyl-modified Belluš−Claisen rearrangement. *Chem. Commun.* **2007**, 1077−1079. (f) Shen, M.-H.; Han, M.; Xu, H.-D. One-Pot Conversion of N-Allyl-α-cyano Esters to α-Allyl-α-cyano Lactams through a Hydrolysis/Ketene Formation/Cyclization/Claisen Rearrangement Sequence. *Org. Lett.* **2016**, *18*, 889−891. (g) Dittrich, N.; Jung, E.-K.; Davidson, S. J.; Barker, D. An acyl-Claisen/Paal-Knorr approach to fully substituted pyrroles. *Tetrahedron* **2016**, *72*, 4676−4689.

(7) (a) Xu, H.-D.; Jia, Z.-H.; Xu, K.; Han, M.; Jiang, S.-N.; Cao, J.; Wang, J.-C.; Shen, M.-H. Copper-Catalyzed Cyclization/aza-Claisen Rearrangement Cascade Initiated by Ketenimine Formation: An Efficient Stereocontrolled Synthesis of α-Allyl Cyclic Amidines. *Angew. Chem., Int. Ed.* **2014**, *53*, 9284−9288. (b) Shen, M.-H.; Xu, K.; Sun, C.-H.; Xu, H.-D. Stereoselective Construction of Bridged trans-aza-Bicyclo[7/6,3/2,1]alkenyl Imines through Ring Expansion aza-Cope Rearrangement. *Org. Lett.* **2015**, *17*, 5598−5601. (c) Chauhan, D. P.; Varma, S. J.; Gudem, M.; Panigrahi, N.; Singh, K.; Hazra, A.; Talukdar, P. Intramolecular cascade rearrangements of enynamine derived ketenimines: access to acyclic and cyclic amidines. *Org. Biomol. Chem.* **2017**, *15*, 4822−4830. (d) Wang, C.-G.; Wu, R.; Li, T.-P.; Jia, T.; Li, Y.; Fang, D.; Chen, X.; Gao, Y.; Ni, H.-L.; Hu, P.; Wang, B.-Q.; Cao, P. Copper(I)-Catalyzed Ketenimine Formation/Aza-Claisen Rearrangement Cascade for Stereoselective Synthesis of α-Allylic Amidines. *Org. Lett.* **2020**, *22*, 3234−3238.

(8) (a) Bowser, A. M.; Madalengoitia, J. S. A 1,3-Diaza-Claisen Rearrangement that Affords Guanidines. *Org. Lett.* **2004**, *6*, 3409−3412. (b) Bowser, A. M.; Madalengoitia, J. S. Synthesis of highly substituted ureas and thioureas through 1,3-diaza-Claisen rearrangements. *Tetrahedron Lett.* **2005**, *46*, 2869−2872. (c) Aranha, R. M.; Bowser, A. M.; Madalengoitia, J. S. Facile 1,3-diaza-Claisen Rearrangements of Tertiary Allylic Amines Bearing an Electron-Deficient Alkene. *Org. Lett.* **2009**, *11*, 575−578. (d) Koya, S.; Yamanoi, K.; Yamasaki, R.; Azumaya, I.; Masu, H.; Saito, S. Selective Synthesis of Eight-Membered Cyclic Ureas by the [6 + 2] Cycloaddition Reaction of 2-Vinylazetidines and Electron-Deficient Isocyanates. *Org. Lett.* **2009**, *11*, 5438−5441. (e) Potter, R. A.; Bowser, A. M.; Yang, Y.; Madalengoitia, J. S.; Ziller, J. W. Structure-Reactivity Relationships of Zwitterionic 1,3-Diaza-Claisen Rearrangements. *J. Org. Chem.* **2013**, *78*, 11772−11782. (f) Walker, J. D.; Madalengoitia, J. S. Optimization of methods for the generation of carbodiimides for zwitterionic 1,3-diaza-Claisen rearrangements. *Tetrahedron Lett.* **2015**, *56*, 3786−3789. (g) Walker, J. D.; Watson, R.; Flemer, S.; Yang, Y.; Madalengoitia, J. S. Intramolecular Formation of Zwitterionic Intermediates in 1,3-Diaza-Claisen Rearrangements. *Org. Lett.* **2017**, *19*, 4010−4013.

(9) (a) Schenck, T. G.; Bosnich, B. Homogeneous Catalysis. Transition-Metal-Catalyzed Claisen Rearrangements. *J. Am. Chem. Soc.* **1985**, *107*, 2058−2066. (b) Li, M.-B.; Wang, Y.; Tian, S.-K. Regioselective and Stereospecific Cross-Coupling of Primary Allylic Amines with Boronic Acids and Boronates through Palladium-Catalyzed C-N Bond Cleavage. *Angew. Chem., Int. Ed.* **2012**, *51*, 2968−2971. (c) Wu, X.-S.; Chen, Y.; Li, M.-B.; Zhou, M.-G.; Tian, S.-K. Direct Substitution of Primary Allylic Amines with Sulfinate Salts. *J. Am. Chem. Soc.* **2012**, *134*, 14694−14697. (d) Li, M.-B.; Li, H.; Wang, J.; Liu, C.-R.; Tian, S.-K. Catalytic stereospecific alkylation of malononitriles with enantioenriched primary allylic amines. *Chem. Commun.* **2013**, *49*, 8190−8192. (e) Wang, Y.; Li, M.; Ma, X.; Liu, C.;

https://dx.doi.org/10.1021/acs.joc.0c02721
J. Org. Chem. 2021, 86, 3065−3073

Gu, Y.; Tian, S.-K. Deammoniative Condensation of Primary Allylic Amines with Nonallylic Amines. *Chin. J. Chem.* **2014**, *32*, 741−751. (f) Wang, Y.; Xu, J.-K.; Gu, Y.; Tian, S.-K. Catalytic stereospecific allylation of protected hydrazines with enantioenriched primary allylic amines. *Org. Chem. Front.* **2014**, *1*, 812−816. (g) Wang, T.-T.; Wang, F.-X.; Yang, F.-L.; Tian, S.-K. Palladium-catalyzed aerobic oxidative coupling of enantioenriched primary allylic amines with sulfonyl hydrazides leading to optically active allylic sulfones. *Chem. Commun.* **2014**, *50*, 3802−3805.

(10) (a) Zhang, J.; Chen, Z.-X.; Du, T.; Li, B.; Gu, Y.; Tian, S.-K. Aryne-Mediated [2,3]-Sigmatropic Rearrangement of Tertiary Allylic Amines. *Org. Lett.* **2016**, *18*, 4872−4875. (b) Gui, Y.; Tian, S.-K. Stereospecific Nucleophilic Substitution of Enantioenriched Tertiary Benzylic Amines via in Situ Activation with Benzyne. *Org. Lett.* **2017**, *19*, 1554−1557. (c) Chen, Z.; Han, L.; Tian, S.-K. Activation and Substitution of 1-Ferrocenylalkylamines with Allenones: Application to Three-Component Synthesis of 4-(1-Ferrocenylalkyl)pyrazoles. *Org. Lett.* **2017**, *19*, 5852−5855. (d) Zhou, M.-G.; Dai, R.-H.; Tian, S.-K. Nucleophilic addition of tertiary propargylic amines to arynes followed by a [2,3]-sigmatropic rearrangement. *Chem. Commun.* **2018**, *54*, 6036−6039. (e) Xu, Y.-N.; Tian, S.-K. Facile construction of three-membered rings via benzyne-promoted Darzens-type reaction of tertiary amines. *Tetrahedron* **2019**, *75*, 1632−1638.

(11) Two examples were reported on the [6 + 2] cycloaddition of a 2-vinylazetidine and allenones in moderate yields. See ref 7d.

(12) Zhao, X.; Liu, D.; Guo, H.; Liu, Y.; Zhang, W. C−N Bond Cleavage of Allylic Amines via Hydrogen Bond Activation with Alcohol Solvents in Pd-Catalyzed Allylic Alkylation of Carbonyl Compounds. *J. Am. Chem. Soc.* **2011**, *133*, 19354−19357.

(13) (a) Dudnik, A. S.; Gevorgyan, V. Metal-Catalyzed [1,2]-Alkyl Shift in Allenyl Ketones: Synthesis of Multisubstituted Furans. *Angew. Chem., Int. Ed.* **2007**, *46*, 5195−5197. (b) Zhang, X.-Y.; Jia, X.-F.; Fang, L.-L.; Liu, N.; Wang, J.-J.; Fan, X.-S. Tandem Reactions of 1,2-Allenic Ketones Leading to Substituted Benzenes and α,β-Unsaturated Nitriles. *Org. Lett.* **2011**, *13*, 5024−5027. (c) Liu, J.; Ma, S. Room temperature Fe(NO3)3·9H2O/TEMPO/NaCl-catalyzed aerobic oxidation of homopropargylic alcohols. *Tetrahedron* **2013**, *69*, 10161−10167.

(14) (a) Klauber, E. G.; Mittal, N.; Shah, T. K.; Seidel, D. A Dual-Catalysis/Anion-Binding Approach to the Kinetic Resolution of Allylic Amines. *Org. Lett.* **2011**, *13*, 2464−2467. (b) Wang, Y.; Xu, Y.-N.; Fang, G.-S.; Kang, H.-J.; Gu, Y.; Tian, S.-K. Kinetic resolution of primary allylic amines via palladium-catalyzed asymmetric allylic alkylation of malononitriles. *Org. Biomol. Chem.* **2015**, *13*, 5367−5371.

(15) Adamson, N. J.; Hull, E.; Malcolmson, S. J. Enantioselective Intermolecular Addition of Aliphatic Amines to Acyclic Dienes with a Pd−PHOX Catalyst. *J. Am. Chem. Soc.* **2017**, *139*, 7180−7183.

(16) Wu, H.; Xie, F.; Wang, Y.; Zhao, X.; Liu, D.; Zhang, W. Pd-catalyzed asymmetric allylic amination using easily accessible metallocenyl P,N-ligands. *Org. Biomol. Chem.* **2015**, *13*, 4248−4254.

(17) Dai, C.; Stephenson, C. R. J. Total Synthesis of Syringolin A. *Org. Lett.* **2010**, *12*, 3453−3455.

(18) Moen, A. R.; Ruud, K.; Anthonsen, T. Combination of stereospecific dihydroxylation and enzyme catalyzed enantioselective resolution for synthesis of enantiopure vicinal diols. *J. Mol. Catal. B: Enzym.* **2008**, *50*, 74−79.

(19) Djouhri-Bouktab, L.; Vidal, N.; Rolain, J. M.; Brunel, J. M. Synthesis of New 3,20-Bispolyaminosteroid Squalamine Analogues and Evaluation of Their Antimicrobial Activities. *J. Med. Chem.* **2011**, *54*, 7417−7421.

https://dx.doi.org/10.1021/acs.joc.0c02721
J. Org. Chem. 2021, 86, 3065−3073